Tangent-groupoid scaling and the quantum construction of Newton motion

Scaling result, 2026-10-30. For a nonnegative scalar potential on the line, coordinate/time dilation has an exact action on the full quantum Hamiltonian, its kernels and its normalized endpoint bridges. With $\hbar = \epsilon S_0$, $t = \epsilon T$ and $x - y = \epsilon v$, the rescaled Hamiltonian has supplied action S_0 and potential $V(q + \epsilon X)$. For $V \in C^3$ with bounded second and third derivatives, its heat kernel has a locally uniform tangent limit and an explicit second-order potential correction. The scale/composition comparison in §7 retains the whole interaction. Normalizing each transition separately generally destroys its semigroup law; normalizing a complete endpoint bridge preserves refinement consistency. The local tangent limit and the scaling that resolves a quantum well retain different information, as §8 makes precise.

State result, 2026-10-27. The tangent groupoid carries a continuous field of observable algebras whose positive-parameter fibers are quantum and whose zero fiber is classical. After fixing an action unit S_0 and setting $\hbar = \epsilon S_0$, the finite quantum products and classical initial-value construction of the quantum product theorem define compatible states on these fibers. On every compact time interval, their values on smooth Weyl observables converge to the classical point state with an explicit $O(\hbar + |\pi|)$ bound, for **every joint schedule** with $\hbar, |\pi| \to 0$.

The result is a state-level limit for the specified coherent preparations and smooth potentials with bounded derivatives of order at least two, including the third derivative used for the rate. The quartic extension in the quantum product theorem uses cutoff Hamiltonians; it does not establish this estimate for uncut quartic quantum evolution. The result neither makes the finite-product propagator uniformly close to the exact quantum propagator as $\hbar \to 0$, nor makes a quantum superposition a single classical trajectory. The groupoid parameter is a geometric deformation parameter until the identification $\hbar = \epsilon S_0$ is chosen. The time mesh is an additional scale.

The wall construction treats boundary domains and impact separately; the whole-line theorem here does not cover that problem. The classical construction and quantum product theorem supply the dynamics used below.

The groupoid and strict quantization are established: in particular, Cariñena, Clemente-Gallardo, Follana, Gracia-Bondía, Rivero and Várilly, Connes' Tangent Groupoid and Strict Quantization (§§3.1–3.3, selected passages read; see the repository companion), already includes the user's work. Landsman–Ramazan's continuous-field construction (abstract) supplies the general Lie-groupoid setting. The following state estimate applies the quantum product theorem's finite-slice bounds to that established structure; no literature priority claim is made.

1. What the parameter joins

For $M = \mathbb{R}$, the tangent groupoid is

$$\mathbb{T}M = (M \times M \times (0, 1]) \;\sqcup\; (TM \times \{0\}). \tag{1}$$

At $\epsilon > 0$, pair arrows compose as $(x, y, \epsilon)(y, z, \epsilon) = (x, z, \epsilon)$; at zero, $(q, v, 0)(q, w, 0) = (q, v + w, 0)$. A local chart near the zero fiber is

$$(q, v, \epsilon) \longmapsto (q + \epsilon v/2,\; q - \epsilon v/2,\; \epsilon). \tag{2}$$

Thus $v = (x - y)/\epsilon$ is a **tangent coordinate**, not automatically a physical velocity. A physical time interpretation requires a separate identification of units and scaling.

The groupoid C^*-algebra is a continuous field with fibers

$$\mathcal{A}_\epsilon \simeq \mathcal{K}(L^2(\mathbb{R})) \quad (\epsilon > 0), \qquad \mathcal{A}_0 \simeq C_0(T^*\mathbb{R}). \tag{3}$$

At zero, Fourier transformation takes tangent-vector convolution to pointwise multiplication of functions of (q,p). The positive fibers are abstractly isomorphic, but the continuous field specifies how families of their elements approach a classical observable.

Choose an action unit $S_0 > 0$ and set

$$\hbar = \epsilon S_0. \tag{4}$$

For $a \in C_c^\infty(T^*\mathbb{R})$, Weyl quantization has kernel

$$K_\epsilon^a(x,y) = \frac{1}{2\pi\epsilon S_0}\int_{\mathbb{R}} e^{i(x-y)p/(\epsilon S_0)} a\Big(\frac{x+y}{2}, p\Big)\, dp. \tag{5}$$

In chart (2),

$$\epsilon K_\epsilon^a(q+\epsilon v/2, q-\epsilon v/2) = \frac{1}{2\pi S_0}\int e^{ivp/S_0} a(q,p) dp. \tag{6}$$

The rescaled kernel is the inverse Fourier transform of the zero-fiber symbol. The established strict-quantization construction makes $\epsilon \mapsto \mathrm{Op}^W_{\epsilon S_0}(a)$, together with a at zero, a continuous section. Equation (6) displays the chart and normalization in this flat case; it is not a new proof of the general continuous-field theorem.

2. The dynamics and the unbounded objects

For the potential and hypotheses of the quantum product theorem define

$$H_\hbar = P^2/(2m) + V(Q), \qquad U_\hbar(t) = e^{-itH_\hbar/\hbar}. \tag{7}$$

The $H_\hbar$ are unbounded operators and the $U_\hbar(t)$ are generally multipliers, not elements of the compact-operator fibers. At $\hbar > 0$ their adjoint action defines automorphisms of the compact operators:

$$\alpha_\hbar^t(A) = U_\hbar(t)^* A U_\hbar(t). \tag{8}$$

The quantum product theorem constructs the unique classical phase-space flow $\Phi_t(q,p)$ before it is used here. On the zero fiber put

$$\alpha_0^t(a) = a \circ \Phi_t. \tag{9}$$

This is classical Hamiltonian evolution with $H_{\mathrm{cl}} = p^2/(2m) + V(q)$.

For $a \in C_c^\infty(T^*\mathbb{R})$, Weyl–Egorov compatibility for the smooth subquadratic potentials of the quantum product theorem gives, for fixed T,

$$\sup_{|t|\le T} \|\alpha_\hbar^t(\mathrm{Op}_\hbar^W(a)) - \mathrm{Op}_\hbar^W(a\circ\Phi_t)\| \le C_{a,T}\hbar^2. \tag{10}$$

Here is the relevant estimate. The finite-time flow has bounded derivatives on compact sets under the stated hypotheses, so $a_s = a \circ \Phi_s$ is smooth, compactly supported, and has uniform symbol bounds for $|s| \le T$. The kinetic commutator is exact. The potential-commutator remainder after removing its classical first-order term has kernel

$$\frac{i}{\hbar}\left[V(x) - V(y) - (x-y)V'\Big(\frac{x+y}{2}\Big)\right] K_\hbar^{a_s}(x,y). \tag{10a}$$

The bracket is bounded by $\|V'''\|_\infty |x-y|^3/24$. Rapid decay of the Fourier transform of a_s in its momentum variable, the change $x - y = \hbar z$, and Schur's test bound the operator norm by $C_{a,T}\hbar^2$. Integrating this commutator remainder by Duhamel's formula gives (10). This uses the standard Weyl calculus and the flow already constructed in the quantum product theorem; it is not being used to prove that flow exists.

3. A quantitative state on every fiber

Use the coherent packet $\psi_\hbar^{q_0,p_0}$ of equation (16) in the quantum product theorem. For a partition π, let $\psi_{\hbar,\pi,j}$ be the state after its first j kick–drift cells, and let $z_{\hbar,\pi}(t_j) = (q_j, p_j)$ be its means. Denote the constructed classical state by $z(t) = \Phi_t(q_0, p_0)$. Define

$$I_a = (2\pi)^{-2} \int_{\mathbb{R}^2} |\widehat{a}(\xi,\eta)|\,(|\xi| + m\omega|\eta|)\,d\xi\,d\eta < \infty. \tag{11}$$

Here $\widehat{a}$ is the ordinary Fourier transform in (q, p). Also put

$$J_a = \frac{1}{2(2\pi)^2} \int_{\mathbb{R}^2} |\widehat{a}(\xi,\eta)|\,(|\xi| + m\omega|\eta|)^2\,d\xi\,d\eta < \infty. \tag{11a}$$

Using the exact Fourier representation of Weyl quantization,

$$\mathrm{Op}_\hbar^W(a) = (2\pi)^{-2} \int \widehat{a}(\xi,\eta) e^{i(\xi Q + \eta P)} d\xi\,d\eta, \tag{12}$$

set $A = \xi(Q - q_j) + \eta(P - p_j)$. The scalar phase $e^{i(\xi q_j + \eta p_j)}$ factors out, $\langle A\rangle_j = 0$, and spectral calculus gives $|\langle e^{iA}\rangle_j - 1| \le \frac{1}{2}\langle A^2\rangle_j$. Consequently,

$$|\langle \mathrm{Op}_\hbar^W(a)\rangle_j - a(q_j, p_j)| \le J_a \left[\mathrm{sd}(Q)_j + \frac{\mathrm{sd}(P)_j}{m\omega}\right]^2. \tag{13}$$

Indeed $\langle A^2\rangle_j = \|A\psi_{\hbar,\pi,j}\|^2$ and the triangle inequality bounds $\|A\psi_{\hbar,\pi,j}\|$ by the weighted sum of the two standard deviations. No Gaussian-shape assumption is needed after the first step. Fourier inversion also gives $|a(z) - a(z')| \le I_a \|z - z'\|_\omega$.

The quantum product theorem's estimates (19) and (26) therefore give, at every partition vertex $t_j \in [0, T]$,

$$\left|\langle \psi_{\hbar,\pi,j}, \mathrm{Op}_\hbar^W(a)\psi_{\hbar,\pi,j}\rangle - a(z(t_j))\right| \le J_a C^2 e^{2\beta T}\hbar + I_a G_T(C_\delta|\pi| + C_\hbar\hbar). \tag{14}$$

The constants $C, \beta, G_T, C_\delta, C_\hbar$ are explicitly defined in that document and depend on the common $\hbar_0, T, V, m, \omega$ and initial data, not on $\hbar$ or the number of cells. No differentiability of an unbounded quantum expectation is used in this step.

With $\hbar = \epsilon S_0$, choose any $\alpha > 0$ and a time unit $T_0 > 0$ so that $|\pi_\epsilon| \le T_0\epsilon^\alpha$. Equation (14) yields

$$\langle \mathrm{Op}_{\epsilon S_0}^W(a)\rangle_{\pi_\epsilon} \longrightarrow a(\Phi_t(q_0, p_0)) \quad \text{at vertices tending to } t, \tag{15}$$

with error $O(\epsilon + \epsilon^\alpha)$ at vertices equal to t. Such a vertex can be inserted in each partition. There is no constraint here on $|\pi_\epsilon|/\hbar$.

This means that the vector states on the positive fibers approach the point-evaluation state on the zero fiber, for the specified family of preparations. For an arbitrary continuous section A, approximate $A(0) \in C_0(T^*\mathbb{R})$ uniformly by compactly supported smooth symbols. Continuity of the field and the norm-one bound on states extend (15) from these symbols to A. At fixed $\hbar$, taking the mesh limit recovers the exact unitary state as in the quantum product theorem. Both routes thus reach the same zero-fiber state; equality or closeness of the two quantum vectors along an arbitrary joint schedule has **not** been established.

4. A precise two-scale test on Euclidean increments

The preceding theorem concerns real-time operator states. It does not claim that Euclidean sample paths are real-time trajectories. The following free Gaussian calculation shows why the groupoid coordinate, the action parameter and the duration must not be conflated.

Consider a drifted free reference increment

$$Y - X = u\delta + \sqrt{\frac{\hbar\delta}{m}}\, Z, \qquad Z \sim N(0,1). \tag{16}$$

Set $\hbar = \epsilon S_0$ and $\delta = T_0\epsilon^\alpha$ with $\alpha > 0$. The physical-time difference quotient is

$$\frac{Y - X}{\delta} = u + \sqrt{\frac{S_0}{mT_0}}\, \epsilon^{(1-\alpha)/2} Z, \tag{17}$$

whereas the tangent-groupoid chart variable is

$$v_\epsilon = \frac{Y - X}{\epsilon} = uT_0\epsilon^{\alpha-1} + \sqrt{\frac{S_0T_0}{m}}\, \epsilon^{(\alpha-1)/2} Z. \tag{18}$$

- If $0 < \alpha < 1$, the physical-time quotient concentrates at u, but v_ϵ is not tight.
- If $\alpha = 1$, v_ϵ converges in law to a nondegenerate Gaussian of mean uT_0 and variance S_0T_0/m.
- If $\alpha > 1$, $v_\epsilon \to 0$ in probability, while the physical-time quotient is not tight.

This is an exact diagnostic for the Euclidean Gaussian reference, not a probability law for real-time Feynman paths. It also does not contradict (15): a groupoid observable is obtained after the Fourier and operator construction, whereas (18) examines a single sampled Euclidean increment.

5. Where blocking sits in the groupoid picture

For each $\epsilon > 0$ the quantum product theorem supplies a complete tower $\rho_{\epsilon,n}$ of normalized conditional densities on dyadic time samples. The relation

$$\rho_{\epsilon,n} = \mathbb{E}_{0,\epsilon}[\rho_{\epsilon,n+1} \mid \mathcal{F}_n] \tag{19}$$

is exact and retains all generated interactions. The integer n counts an integration/coarse-graining operation; ϵ labels the deformation field (4). The conditional maps are generally not invertible, so they are not automatically arrows of the tangent groupoid.

Equation (15) connects these scales for the real-time **observable states** even when they are tied by a single chosen schedule. It does not follow from the $1/Z_\epsilon$ Euclidean bound, which need not be uniform in ϵ, and it does not promote Euclidean minimization into real-time Newton dynamics.

The zoom action on the tangent groupoid is a geometric rescaling of near-diagonal arrows. It is useful for organizing a comparison with RG, but it is not by itself an integration of fluctuations or a theorem that the physical action constant runs. Likewise, the real-time Hamilton principal action composes by stationary elimination on a specified noncaustic branch; infimal convolution is appropriate for a genuine minimizing Euclidean problem or a separately proved special case. The kernel calculation below makes this distinction explicit.

6. Regulators and a normalized blocking test

The connection is already part of Rivero's work. The appendix to his 1997 tangent-groupoid note (pp. 7–8, passages read) constructs a **toy RG triangle**: a dilation rescales a bare sequence approaching the tangent boundary to a chosen nonzero reference slice, with limits at different slices related by rescalings. His 1998 Feynman-formula proposal (pp. 2–3, equations (5)–(10), passages read) similarly distinguishes two regulators and calls its controlled objects bare, renormalized and dressed series. Its control transformation is left unspecified; its control constant is

arbitrary. The 1999 strict-quantization paper with Rivero and coauthors supplies the groupoid quantization framework. These are the provenance of the present question, not a claim that the recent calculation invented the groupoid/RG connection.

The different appearances of epsilon need an explicit dictionary. Let ϵ_G be a dimensionless groupoid deformation parameter; in the specified state formulation, choosing $\epsilon_G = \hbar/S_0$ assigns the dimensionful phase constant to that parameter. Let $\eta = |\pi|/T_0$ be a dimensionless mesh scale, and let α_π be an action-dimensional bare phase regulator. A possible control map, already recorded in the 1998 test, is

$$S_\pi^{\mathrm{bare}} = Z_S(\eta) S_\pi, \qquad \hbar_{\mathrm{eff}} = \frac{\alpha_\pi}{Z_S(\eta)}.$$

For example, the *specified* choices $Z_S(\eta) = \eta$ and $\alpha_\pi = \eta\hbar$ give $e^{iS_\pi^{\mathrm{bare}}/\alpha_\pi} = e^{iS_\pi/\hbar}$ at every mesh. This algebra illustrates how a vanishing bare regulator can coexist with a fixed positive effective action; substituting the same regulator into an unscaled action would give a different phase.

An exact Gaussian blocking test. Let $m > 0$, $u \in \mathbb{R}$ and let $a \geq 0$ be an action-dimensional parameter. Define a normalized transition kernel by

$$K^u_{a,t}(x, dy) = \mathcal{N}(x + ut, at/m)(dy), \qquad t > 0,$$

with zero variance interpreted as a Dirac measure. For $a = \hbar > 0$ and $u = 0$ this is the free Euclidean quantum kernel; arbitrary u gives the drifted Gaussian reference, not a real-time path measure. Integrating out an intermediate position preserves a exactly:

$$K^u_{a,t} K^u_{a,v} = K^u_{a,t+v}.$$

If time and position are then rescaled by $t_{\mathrm{old}} = r t_{\mathrm{new}}$, $x_{\mathrm{old}} = s x_{\mathrm{new}}$ with $r, s > 0$, the normalized kernel obeys

$$K^u_{a,rt}(sx, sB) = K^{ru/s}_{ra/s^2,t}(x, B) \quad \text{for every Borel set } B.$$

These identities follow from adding independent Gaussian increments and rescaling their means and variances; they include $a = 0$. Under the diffusive choice $r = b$, $s = \sqrt{b}$, the action coefficient stays fixed. When $u = 0$, every $a \geq 0$ is a fixed point of this particular normalized block-and-rescale map. At $a = 0$, retaining u gives the deterministic free Newton motion $x(t) = x(0) + ut$. Thus this exact free RG control neither selects a positive action nor excludes the classical branch. It is an illustrative Gaussian comparison, not a result about all interacting blocking maps or complete record instruments; compare the free increment and the action-normalization identity.

The two-regulator bridge and a residual action. For the free Gaussian endpoint bridge on a partition with $d = N - 1$ interior points, whitening the quadratic action gives $2\Delta S/a \sim \chi^2_d$ and hence

$$\mathbb{E}\Delta S = \frac{da}{2}, \qquad \mathrm{Var}(\Delta S) = \frac{da^2}{2}.$$

If the maximum cell length and a_N tend to zero while $da_N \to \ell$, the bridges can be coupled to converge uniformly almost surely to the classical chord, while $\Delta S \to \ell/2$ in L^2. This uses one Brownian bridge and its polygonal interpolations. For equal cells $\tau = T/N$ and $a_N = b\tau$, the supplied b has energy units and $\ell = bT$. The positive residual is selected by the scaling; at every finite mesh the action distribution has values arbitrarily near zero.

The separately controlled reference family fixes $m_R, \kappa_R > 0$ and chooses $m_B = \alpha_\pi m_R/\kappa_R$, so its normalized free kernels are preserved as $\alpha_\pi \to 0$; the bare mass then tends to zero. This is different from the preceding fixed-mass limit and does not derive the reference action. The regulator manuscript, Proposition 2 and its running-coefficient section, contains the full arbitrary-partition proof and source reading; the PDF, B08 and its derivation retain the detailed record. The manuscript also gives a bounded-speed shrinking sine example with a fixed action defect and growing acceleration, and a separate scalar stationary-phase/delta-weight calculation.

Those examples motivate testing which independent force or derivative bounds suppress such a defect; they do not make the defect universal or identify it with $\hbar$.

The interacting scalar comparison in §7 now gives an exact dilation map commuting with kernel composition and endpoint-bridge marginalization. It uses an already defined Hamiltonian and retains its complete kernels; it supplies no closed finite list of effective couplings. The wider RG question is uniform control of generated interactions at the required physical times and preparations, and any independent universal calibration of the action. Geometric zoom, coarse-graining and running parameters can participate in that construction; they remain different operations. The corrected groupoid comparison and scale correspondence keep these claims and their corrections distinct.

Geometric zoom and kernel composition

The adiabatic zoom acts by $(x, y, \epsilon) \mapsto (x, y, \lambda^{-1}\epsilon)$ and $(x, v, 0) \mapsto (x, \lambda v, 0)$. Debord–Skandalis 2014 and van Erp–Yuncken 2019 (§1.1 and §4.1, Definition 17, passages read; Debord–Skandalis metadata only) relate this geometry to pseudodifferential kernels homogeneous modulo smoothing terms; the zero-fiber restriction gives their principal symbols. This source result identifies geometric structure, not a Wilsonian integration or a law for physical $\hbar$. Earlier identifications of all three were too broad.

The user's 2003 note (Rivero, "Flashes of noncommutativity", arXiv:math/0302285, full read) introduces the groupoid over configuration space with arrows (x, y, u), u an elapsed time, and law $(x, y, u) \circ (y, z, v) = (x, z, u + v)$. Its convolution algebra has product $(AB)(x, z, t) = \iint A(x, y, r)\, B(y, z, t - r)\, dy\, dr$; Fourier transformation in t gives, with $\varepsilon \sim 1/\hat{t}$, the product of the $\varepsilon > 0$ part of the tangent groupoid, so the latter's algebra sits inside the former's (the 2003 note's second flash). Refining a trajectory is factoring an arrow (x, z, u) through intermediate points and times, which is Newton's insertion of $t_{2.6}$.

For endpoint functions define min-plus convolution by

$$(A \otimes B)(x, z) := \inf_y \{A(x, y) + B(y, z)\}.$$

It eliminates the intermediate endpoint; it does not add the action on each individual composable pair of arrows.

Proposition 1. For $M > 0$, constant force F, and $u > 0$, set

$$S_u(x, y) = \frac{M(y - x)^2}{2u} + \frac{Fu}{2}(x + y) - \frac{F^2u^3}{24M}.$$

Then $(S_u)_{u>0}$ is a semigroup of kernels under min-plus convolution: $S_r \otimes S_v = S_{r+v}$ for $r, v > 0$. This is the corrected action ℓ_u of the refinement note, equation (2). The corresponding constant-force unitary propagators satisfy $U_{r+v} = U_r U_v$; their kernels compose by integration over the intermediate position, in the oscillatory/distributional sense where the real-time integrals are not absolutely convergent.

Proof. The sum $S_r(x, y) + S_v(y, z)$ is a strictly convex quadratic in y with unique minimizer

$$y_* = \frac{vx + rz}{r + v} - \frac{Frv}{2M}.$$

Substitution gives $S_{r+v}(x, z)$. Equivalently, y_* is the value at time r of the action-minimizing constant-force path joining x to z. The unitary law follows from the self-adjoint Hamiltonian; its kernel statement is operator composition. □

The min-plus law is not an ordinary groupoid morphism. For free motion with $M = 1$, $S_1(0, 1) + S_1(1, 0) = 1$, whereas $S_2(0, 0) = 0$. The equality holds only after minimizing over the intermediate point ($y = 0$ in this example).

The displayed S is the exact discrete Lagrangian of discrete mechanics (Marsden and West 2001, metadata), and discrete Lagrangian mechanics on groupoids is Weinstein's (1996) (metadata). The kick–drift–kick cell $S_\delta = K_{F\delta/2} D_\delta K_{F\delta/2}$ of the refinement note is the variational (Störmer–Verlet) integrator of the trapezoidal discrete Lagrangian L_δ, and the cubic counterterm turns L_δ into the exact one. The laws of Proposition 1 can be related through a Euclidean Laplace limit: under coercivity and regularity conditions, $-\hbar \log \int e^{-S/\hbar} dy \to \inf_y S$ as $\hbar \to 0$ (Litvinov 2005, metadata). This does not turn an oscillatory real-time integral into a probability integral, nor make every stationary real-time branch a global minimum.

7. Interacting kernels under tangent scaling

Here the kinetic energy, mass and potential are specified. The conclusions do not follow from the groupoid law without those dynamical inputs. Work on $\mathbb{R}$, with $m, S_0 > 0$ and $V \geq 0$. For the estimates assume $V \in C^3$ and bounded V'' and V'''. Use the nonnegative Friedrichs Hamiltonians

$$H_{a,V} = -\frac{a^2}{2m}\partial_x^2 + V(x), \qquad a > 0.$$

Their Euclidean kernels $K_{a,V}(t; x, y)$ are those of $e^{-tH_{a,V}/a}$. The action-dimensional a is a family parameter; the semiclassical family below sets $a = \hbar = \epsilon S_0$. Neither ϵ nor a duration is being assigned the units of physical $\hbar$.

Exact dilation and composition

For dimensionless $r, s > 0$ and a fixed centre q, put

$$(D_{q,s} f)(X) = s^{1/2} f(q + sX), \qquad a' = \frac{ra}{s^2}, \qquad V_{r,s,q}(X) = \frac{r^2}{s^2} V(q + sX). \tag{20}$$

Direct differentiation gives the exact operator identity

$$\frac{r}{a} D_{q,s} H_{a,V} D_{q,s}^{-1} = \frac{1}{a'} H_{a', V_{r,s,q}}. \tag{21}$$

Unitary conjugation preserves the Friedrichs domains and proves (21) also for the closed forms. Spectral calculus therefore gives

$$s K_{a,V}(rt; q + sX, q + sY) = K_{a', V_{r,s,q}}(t; X, Y). \tag{22}$$

The same conjugation holds for real-time unitary propagators; their kernels need not be positive functions. We use (22) as a probability calculation only in Euclidean time.

Keeping m fixed in (20) is a declared normalization, not an invariant meaning of action running. Equivalently keep a fixed, use $m' = ms^2/r$ and $\widehat{V}(X) = rV(q + sX)$, and obtain the same rescaled kernel from $H_{a,m',\widehat{V}}$. Here the extra mass subscript makes the kinetic coefficient explicit. Physical calibration must distinguish these parametrizations; the coordinate/time identity alone cannot do so.

Composition commutes with this dilation. Indeed, substituting $z = q + sZ$ in $K(r(t+u)) = \int K(rt) K(ru)\, dz$ cancels exactly one kernel Jacobian and gives the composition law for the right side of (22). For an endpoint bridge with N cells, the N kernel factors contribute s^{-N}, its $N-1$ internal integrations contribute s^{N-1}, and division by the total endpoint kernel contributes s. Their product is one. Thus the normalized bridge and every marginal transform exactly, before or after eliminating any subset of inserted points. No generated term is dropped. This extends the free comparison in §6 to a specified interacting Hamiltonian; the dilation itself is invertible, whereas eliminating bridge points generally is not.

Inserted points and the boundary product

The user's insertion question distinguishes the arrow law from the kernel law. At fixed $\epsilon > 0$, composable arrows have exactly $(x-y)+(y-z) = x-z$. In symmetric chart coordinates write $u = (x-y)/\epsilon$, $v-u = (y-z)/\epsilon$ and $q = (x+z)/2$. The two factors have different centres:

$$q_1 = \frac{x+y}{2} = q + \frac{\epsilon}{2}(v-u), \qquad q_2 = \frac{y+z}{2} = q - \frac{\epsilon}{2}u. \tag{22a}$$

For kernels A_ϵ, B_ϵ with absolutely convergent product, put $\kappa_\epsilon^A(q,v) = \epsilon A_\epsilon(q+\epsilon v/2, q-\epsilon v/2)$, and similarly for B. Substitution of $y = x - \epsilon u$ in the product integral gives

$$\kappa_\epsilon^{AB}(q,v) = \int \kappa_\epsilon^A\big(q + \tfrac{\epsilon}{2}(v-u), u\big)\, \kappa_\epsilon^B\big(q - \tfrac{\epsilon}{2}u, v-u\big)\, du. \tag{22b}$$

The tangent-bundle law at zero still permits the decomposition $v = u + (v-u)$, but the separate finite-scale centres have collapsed to the same q. Its convolution is consequently fibrewise; Fourier transformation makes its product commutative. The exact positive-scale formula retains the shifts, rather than merely adding two tangent vectors at an unchanged base point. Thus the bare zero-fibre bundle does not retain where an intermediate finite-scale point was inserted.

For heat kernels with durations $t, u > 0$, refinement additionally requires integrating $K(t;x,y)K(u;y,z)$ over that inserted point to obtain $K(t+u;x,z)$. The product at one chosen y is one integrand, not the whole composed kernel. A duration split is extra data beyond the ordinary pair-groupoid arrow; the time-graded formulation in §6 records it. This is compatible with (22b) and with the distinction between elimination and pointwise action addition already proved there.

The matched tangent limit

Take $a = \epsilon S_0$ and $r = s = \epsilon$. Equations (20)–(22) become

$$\widetilde{H}_{\epsilon,q} = -\frac{S_0^2}{2m}\partial_X^2 + V(q+\epsilon X), \tag{23a}$$

$$\epsilon K_{\epsilon S_0, V}(\epsilon T; q+\epsilon X, q+\epsilon Y) = K_{S_0, V(q+\epsilon\,\cdot)}(T; X, Y). \tag{23b}$$

The potential on the right tends locally to $V(q)$. On C_c^∞, the operators consequently converge to $H_{0,q} = -S_0^2\partial_X^2/(2m) + V(q)$ in norm on each test vector. Fix an energy $E_{\mathrm{ref}} > 0$. For f in this core,

$$\begin{aligned}(\widetilde{H}_{\epsilon,q} + E_{\mathrm{ref}})^{-1}(H_{0,q} + E_{\mathrm{ref}})f - f \\ = (\widetilde{H}_{\epsilon,q} + E_{\mathrm{ref}})^{-1}(H_{0,q} - \widetilde{H}_{\epsilon,q})f \longrightarrow 0.\end{aligned}$$

The resolvents have norm at most E_{ref}^{-1} and $(H_{0,q}+E_{\mathrm{ref}})C_c^\infty$ is dense, so convergence extends strongly to every vector. Continuous functional calculus gives strong convergence of the heat operators. Local kernel convergence, with a rate, follows independently from the bridge calculation below. There is no claim of operator-norm convergence on this one fixed L^2 space: for a harmonic potential every positive-ϵ heat operator is compact, while the limiting free heat operator times $e^{-TV(q)/S_0}$ is not. A norm limit of compact operators would be compact. This comparison under dilation is distinct from continuity of the groupoid field with different zero and positive fibers.

For the symmetric chart $X = v/2$, $Y = -v/2$, define

$$\kappa_\epsilon(q,v;T) = \epsilon K_{\epsilon S_0, V}(\epsilon T; q+\epsilon v/2, q-\epsilon v/2). \tag{24}$$

Then, locally uniformly in q, v and in T bounded away from zero and infinity,

$$\kappa_\epsilon(q,v;T) \longrightarrow \kappa_0(q,v;T) = \sqrt{\frac{m}{2\pi S_0 T}}\exp\left[-\frac{mv^2}{2S_0T} - \frac{TV(q)}{S_0}\right]. \tag{25}$$

The tangent Fourier transform is exactly

$$\int e^{-ivp/S_0}\kappa_0(q,v;T)\,dv = \exp\left[-\frac{T}{S_0}\left(\frac{p^2}{2m}+V(q)\right)\right]. \tag{26}$$

At fixed q, tangent-vector convolution of these Gaussian kernels adds T; after Fourier transformation this is multiplication of the displayed classical heat symbols. The boundary value is the semiclassical symbol of the specified family $H_{\epsilon S_0,V}$. It is not the ordinary principal symbol of a fixed second-order operator, for which V is lower order. Nor is (26) a real-time path probability or a claim that every heat operator lies in a compact-operator fiber. For example, a constant potential gives bounded multipliers, not elements of $\mathcal{K}(L^2(\mathbb{R}))$, and its symbol does not vanish at spatial infinity.

This scaling follows a family of near-diagonal arrows as well as changing the physical duration. It is not a single orbit of the adiabatic zoom in §6, which fixes the positive-fiber endpoints. The kinetic variance $\hbar t/m = \epsilon^2 S_0 T/m$ explains why the tangent displacement is order ϵ here. At fixed positive physical $\hbar$, a short-time diffusive displacement instead has order $\sqrt{t}$.

The first surviving potential correction

Write B_s, $0 \le s \le 1$, for a standard real Brownian bridge and put

$$\sigma^2 = \frac{S_0 T}{m}, \qquad \xi_s = (\tfrac{1}{2}-s)v + \sigma B_s, \qquad A_\epsilon = \frac{T}{S_0}\int_0^1 V(q+\epsilon\xi_s)\,ds.$$

The Feynman–Kac bridge formula gives

$$\kappa_\epsilon = \sqrt{\frac{m}{2\pi S_0 T}}e^{-mv^2/(2S_0T)}\mathbb{E}e^{-A_\epsilon}. \tag{27}$$

The formula is established: Simon 1999, Theorem 1.1 and equations (1.1)–(1.3), printed pp. 1–2, passages read. The Brownian-bridge expansion is also standard Wigner–Kirkwood machinery; see Jizba–Zatloukal 2014, §IV, equations (34)–(42), passages read. The constants and local remainder used here are derived next; no novelty is asserted for the method.

Let $I_1 = \int_0^1 \xi_s ds$ and $I_2 = \int_0^1 \xi_s^2 ds$. The bridge covariance $\min(s,u) - su$ gives

$$\mathbb{E}I_1 = 0, \qquad \mathbb{E}I_1^2 = \frac{\sigma^2}{12}, \qquad \mathbb{E}I_2 = \frac{v^2}{12}+\frac{\sigma^2}{6}.$$

For example, integrating the covariance first in u gives $s(1-s)/2$, and its integral in s is $1/12$. Differentiating $\mathbb{E}e^{-A_\epsilon}$ twice at zero therefore yields

$$\kappa_\epsilon = \kappa_0[1+\epsilon^2 C(q,v,T)+O(\epsilon^3)], \tag{28a}$$

$$C = \frac{T^3[V'(q)]^2}{24mS_0} - \frac{TV''(q)v^2}{24S_0} - \frac{T^2V''(q)}{12m}. \tag{28b}$$

There is no linear term in this symmetric endpoint kernel. To justify the expansion, $V \ge 0$ bounds e^{-A_θ} by one for real θ. Bounded V'' gives $|V'(q+\theta\xi)| \le |V'(q)| + \|V''\|_\infty|\theta\xi|$. The first three derivatives of e^{-A_θ} are consequently bounded by polynomials in $\int|\xi_s|^j ds$, $j \le 3$, with locally uniform coefficients for $|\theta| \le 1$. All required Gaussian moments are finite. For an explicit remainder bound put $\beta = T/S_0$, $G = |V'(q)|$, $L = \|V''\|_\infty$, $M = \|V'''\|_\infty$ and $U_j = \int_0^1|\xi_s|^j ds$. Define

$$\mathcal{M}(q,v,T) = \mathbb{E}\big[\beta^3(GU_1+LU_2)^3 + 3\beta^2 L(GU_1+LU_2)U_2 + \beta M U_3\big].$$

The absolute value of the relative remainder in (28a) is at most $\epsilon^3 e^{\beta V(q)}\mathcal{M}(q,v,T)/6$, for $0 \le \epsilon \le 1$. This follows from $|\partial_\theta^3 e^{-A_\theta}| \le |A'_\theta|^3 + 3|A'_\theta A''_\theta| + |A'''_\theta|$ and the bounds above. The Gaussian moment constant is finite and locally uniform. There is no uniform assertion over all q, all durations or arbitrary singular potentials.

Normalization that survives refinement

The full normalized bridge on the rescaled path ξ has density

$$R_\epsilon = \frac{e^{-A_\epsilon}}{\mathbb{E}e^{-A_\epsilon}} = 1 - \epsilon\frac{TV'(q)}{S_0}I_1 + O_{L^1}(\epsilon^2) \tag{29}$$

relative to its free bridge. The same differentiation bounds prove this L^1 expansion, with its normalizer bounded away from zero on the stated compact sets. Thus the bridge changes at first order even though the integrated symmetric kernel changes only at second order. In physical coordinates its mean is

$$\mathbb{E}[q + \epsilon\xi_s] = q + \epsilon(\tfrac{1}{2} - s)v - \epsilon^2\frac{T^2V'(q)}{2m}s(1-s) + O(\epsilon^3). \tag{30}$$

Here the expectation is under the interacting bridge. The covariance $\mathrm{Cov}(\xi_s, I_1) = \sigma^2 s(1-s)/2$ proves the correction; Gaussian moments justify the same expansion with the inserted ξ_s. Its sign agrees with the Euclidean stationary equation $m\ddot{q} = +V'(q)$, and must not be substituted for the real-time Newton force.

This comparison is uniform over refinements at the level of lifted densities. On any partition of $[0,1]$, replace the integral in A_ϵ by the endpoint trapezoidal sum and call it $A_{\epsilon,\pi}$. For every continuous bridge path,

$$\sup_\pi \left| A_{\epsilon,\pi} - \frac{TV(q)}{S_0} \right| \le \frac{T}{S_0}\sup_s |V(q+\epsilon\xi_s) - V(q)| \longrightarrow 0.$$

All weights lie in $[0,1]$, so dominated convergence gives $\sup_\pi \|R_{\epsilon,\pi} - 1\|_{L^1} \to 0$. At fixed ϵ, ordinary mesh refinement gives $R_{\epsilon,\pi} \to R_\epsilon$ in L^1. Free polygonal interpolation converges uniformly pathwise as $|\pi| \to 0$. Therefore any joint schedule $\epsilon \to 0$, $|\pi| \to 0$ has the free rescaled bridge as its weak path limit. No relation between the dimensionless mesh and ϵ is required. Total variation is asserted for densities lifted to the common free path space, not for polygon-supported measures against a non-polygonal Brownian law. Exact projective consistency belongs to the complete heat-kernel bridges; trapezoidal single-cell weights acquire generated terms when blocked.

Dividing each transition by its row integral is a different operation. For $V(x) = m\omega^2x^2/2$, the Mehler kernel is

$$K_{a,V}(t;x,y) = \sqrt{\frac{m\omega}{2\pi a\sinh\omega t}}\exp\left[-\frac{m\omega[(x^2+y^2)\cosh\omega t - 2xy]}{2a\sinh\omega t}\right].$$

It solves the heat equation with the delta initial condition. Completing the square shows that its row-normalized transition is

$$N_t(x,dy) = \mathcal{N}\left(\frac{x}{\cosh\omega t}, \frac{a}{m\omega}\tanh\omega t\right)(dy). \tag{31}$$

The mean of N_tN_u is $x/(\cosh\omega t\cosh\omega u)$, whereas that of N_{t+u} is $x/\cosh\omega(t+u)$. For $x \neq 0$, $t, u > 0$ they differ. Hence this normalized family is not a semigroup. A genuine Markov semigroup, when a positive ground state exists, is the ground-state transform, which retains the eigenfunction ratios and vacuum-energy factor.

8. Which zoom resolves a quantum well?

The ordinary tangent chart samples distances of order ϵ and, in (23), durations of order ϵ. It freezes the potential locally. To resolve a well rather than a diagonal kernel, use a different scaling. For the separate model $V(x) = \lambda x^{2k}$, $k \ge 1$, $\lambda > 0$, set $\hbar = \epsilon S_0$, $q = 0$ and

$$s = \epsilon^{1/(k+1)}, \qquad r = \epsilon^{(1-k)/(k+1)}. \tag{32}$$

Equations (20)–(22) now give $a' = S_0$ and $V_{r,s,0}(X) = \lambda X^{2k}$ exactly. Equivalently,

$$D_{0,s} H_{\epsilon S_0, V} D_{0,s}^{-1} = \epsilon^{2k/(k+1)} H_{S_0,V}. \tag{33}$$

The equal exponents follow by balancing $\epsilon^2/s^2 = s^{2k}$. This is the standard homogeneous spectral dilation, not a new gap theorem; the quartic construction already gives its $k = 2$ case, and matrix-model scaling has its own operator and invariant-sector conditions.

Limit, with S_0 fixed	Position scale	Physical duration scale	What remains
Matched ordinary tangent kernel	ϵ	ϵ	Free tangent heat kernel times $e^{-TV(q)/S_0}$
Harmonic well, $k = 1$	$\epsilon^{1/2}$	1	Full fixed-action harmonic dynamics
Quartic well, $k = 2$	$\epsilon^{1/3}$	$\epsilon^{-1/3}$	Full fixed-action quartic dynamics

The durations multiply a fixed reference time, and the position factors multiply a fixed reference length or the coordinate X in (20). The two monomial gaps scale respectively as ϵ and $\epsilon^{4/3}$ by (33); their existence is supplied by the confining Hamiltonians. The growing quartic duration lies outside the bounded-T local estimate (28). A well-resolving quantum limit therefore cannot be inferred from that ordinary tangent estimate. It is a different rescaled family; it does not replace the commutative zero fiber of (3).

All these constructions admit any supplied $S_0 > 0$. They construct no independent action floor, and no preferred dimensional normalization of the groupoid parameter. Identifying a particular member with measured physical $\hbar$ still requires physical information. Normalized free refinement at action zero remains the classical countermodel in §6.

9. Consequence for STATE

The constructed $0 + 1$ quantum family, the semiclassical Newton flow and the user's tangent-groupoid framework meet in a state theorem with a joint mesh/action error bound. The scalar scaling question now also has an exact dilation/composition map, a controlled local interacting heat kernel and a refinement-compatible normalized bridge. Ordinary tangent and well-resolving limits use different spatial and temporal scales. The strong product limit remains a fixed-$\hbar$ theorem. Physical calibration, uniform control over growing times or noncompact preparations, and four-dimensional Yang–Mills construction remain separate obligations.